

DBSCAN (**Density-Based Spatial Clustering of Applications with Noise**) is a clustering algorithm that groups together points that are **closely packed** and marks points in low-density regions as **outliers (noise)**.

Unlike K-Means:

K-Means	DBSCAN
Need to specify K	No need to specify number of clusters
Assumes spherical clusters	Can find arbitrary-shaped clusters
Every point belongs to a cluster	Can identify noise/outliers
Sensitive to outliers	Naturally handles outliers

Real-Life Analogy

Imagine you're looking at people in a park.

Dense Areas

- A group of people standing together near a food stall.
- Another group sitting together near a fountain.

These become clusters.

Isolated People

- One person walking alone.
- One person sitting far away.

These become noise (outliers).

DBSCAN works exactly this way.

Two Important Parameters

1. ϵ (Epsilon)

Maximum distance within which points are considered neighbors.

Think:

"Who are my nearby friends?"

Suppose $\epsilon = 2$ meters.

Anyone within 2 meters is considered a neighbor.

2. MinPts

Minimum number of neighbors required to form a cluster.

Example:

MinPts = 3

A point needs at least 3 nearby points to be considered part of a dense region.

Example Dataset

Let's take points:

A (1, 1)
B (1, 2)
C (2, 1)
D (2, 2)

E (8, 8)
F (8, 9)
G (9, 8)
H (9, 9)

I (15, 15)

Visual representation:

(1, 1) A B (1, 2)
(2, 1) C D (2, 2)

E F
G H

I

Assume:

$\varepsilon = 2$

MinPts = 3

Step 1: Find Neighbors

For point A:

Nearby points:

A
B
C
D

Total = 4 points

Since:

$4 \geq \text{MinPts}(3)$

A becomes a Core Point.

For point E:

Nearby points:

E
F
G
H

Total = 4

Again:

$4 \geq 3$

E becomes Core Point.

For point I:

Nearby points:

I only

Total = 1

Since:

$1 < 3$

Not enough density.

I becomes Noise.

Types of Points in DBSCAN

Core Point

Has enough neighbors.

Example:

A
B
C
D
E
F
G
H

All are core points.

Border Point

Not enough neighbors itself, but reachable from a core point.

Example:

Core ---- Border

Border gets assigned to cluster.

Noise Point

Not reachable from any dense region.

Example:

I (15, 15)

Noise.

Building Clusters

Start from A.

A → B → C → D

All density-connected.

Cluster 1:

{A, B, C, D}

Start from E.

E → F → G → H

Cluster 2:

{E, F, G, H}

Point I:

Noise

Final Result

Cluster 1:

A B C D

Cluster 2:

E F G H

Noise:

I

Example with Strange Shape

Suppose data looks like a crescent moon:

```
*****
  *
   *
    *
   *****
```

K-Means tries to split using circles:

Wrong clusters

DBSCAN follows density:

Entire crescent = one cluster

This is one of DBSCAN's biggest strengths.

Why It Is Called Density-Based?

Instead of asking:

"How far is this point from a centroid?"

DBSCAN asks:

"Are there enough neighboring points around me?"

If yes → cluster.

If no → noise.

Interview Question

Why would you choose DBSCAN over K-Means?

Answer:

1. Number of clusters is unknown.
 2. Dataset contains outliers.
 3. Clusters have irregular shapes.
 4. Need automatic noise detection.
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Simple Shopping Example

Imagine customer spending data:

Customer A = ₹5000

Customer B = ₹5200

Customer C = ₹5100

Customer D = ₹4950

Customer E = ₹25000

Customer F = ₹26000

Customer G = ₹25500

Customer H = ₹200000

DBSCAN might discover:

Cluster 1:

Regular customers

Cluster 2:

Premium customers

Noise:

Customer H

Customer H is so different that DBSCAN flags it as an outlier automatically.

One-line memory trick

K-Means = "Find centers and assign points."

DBSCAN = "Find dense neighborhoods and grow clusters."